

Introduction to Microeconomics – S102015CR

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Problem Set 1

I. Chapter 1. Methodology

Exercise 1

Which of the following statements is correct? Select one

- a. The opportunity cost is a concept that measures direct monetary costs.
- b. Iris hesitates between self-driving and public transportation for a holiday travel. The opportunity cost of self-driving is equal to the cost of public transportation.
- c. The opportunity cost accounts for alternative uses of resources such as time and money.
- d. The opportunity cost of buying a property is always smaller than the local rental rate for a comparable house.

Exercise 2

Which of the following statements is correct? Select one

- a. The opportunity cost of leisure includes the wage that could be earned by working instead.
- b. The opportunity cost of attending university includes the future pension that the individual will obtain upon retiring.
- c. The opportunity cost of buying a new car is always smaller than the local rental rate for a comparable car.
- d. The opportunity cost of pursuing a university degree is higher in recessions (when the unemployment rate rises) than in economic booms.

Exercise 3

Decide whether the following topics are part of micro- or macroeconomics:

- 1. A family choosing how much to save.
- 2. Impact of public regulation on car pollution.
- 3. Effect of an increase of the savings rate on economic growth.
- 4. A company choosing the number of its employees.
- 5. The relationship between the inflation rate and the variation in money supply.

Exercise 4

On average, pupils do better at school (have better test results) if they study in smaller classes with fewer classmates and are taught by teachers that are paid more. Which of the following conclusions is correct?

- a. There exists a positive correlation between the salaries of teachers and the test results of the pupils as well as a negative correlation between class size and the test results of the pupils.
- b. The observation proves that the salaries of teachers and class size influence student performance.
- c. The observation proves that the best teachers, who are also paid the most, prefer working in smaller classes.
- d. The observation proves nothing.

II. Chapter 2. Markets

Exercise 5

Why do brown eggs cost more than conventional ones (white eggs)? Hint: consider Figure 1a and Figure 1b below, which plot the demand (in blue) and supply (in red) curves of eggs

Figure 1a

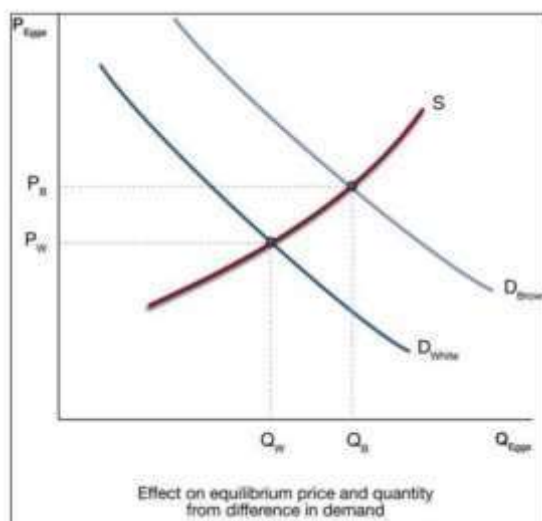
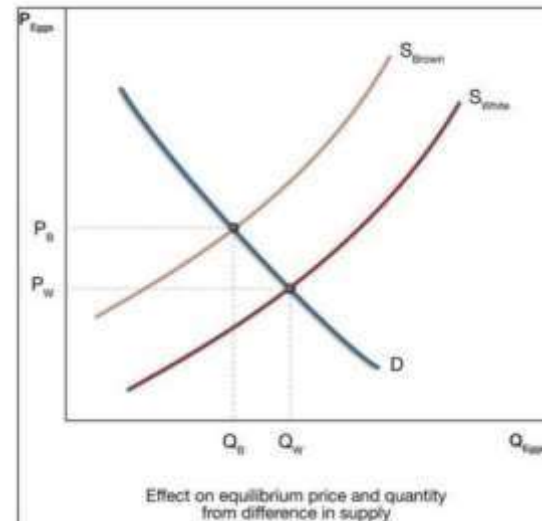


Figure 1b



Exercise 6

Using a supply-and-demand diagram, answer the following questions. First, why do the prices of roses increase right before Valentine's Day?

- A. This outcome is the result of a move of the equilibrium price and quantity along the supply curve.
- B. This outcome is the result of an outward shift of the demand curve.
- C. This outcome is the result of a shift of the supply curve.
- D. This outcome is the result of a shift of both demand and supply.

Exercise

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Consider the following supply and demand functions:

$$Q^S(P) = P - 1 \text{ and } Q^D(P) = 5 - P$$

Where Q^S is the amount of a good supplied on the market, Q^D is the amount of a good demanded on the market and P is the price of this good in this market.

Assume that due to an increase in production costs the supply function has changed as follows:

$$Q^S(P) = P - 3$$

It follows that after the increase in production costs: a. the equilibrium price has increased from 1 to 3.

b. producers' revenue has increased from 3 to 4.

c. the quantity traded on the market has decreased from 2 to 1.

d. total spending on this good has increased from 4 to 6

Exercise 8

The market for wheat is initially in equilibrium. What are the overall consequences of a negative shock to supply caused by heavy rains and of an increase of the price of corn (which is considered to be a substitute for wheat)?

a. The price of wheat increases but the effect on the quantity exchanged is undetermined.

b. The price of wheat and the quantity exchanged increase.

c. The price of wheat increases and the quantity exchanged decreases.

d. Nothing can be said about the direction of the changes in the quantity traded and in price.